

$$K_A < K_B$$

- 13** Diagram 1 shows a ripple tank experiment in which plane waves are diffracted through a narrow slit in a metal sheet.

Diagram 2 shows the same tank with a slit of greater width.

In each case, the pattern of the waves incident on the slit and the emergent pattern are shown.

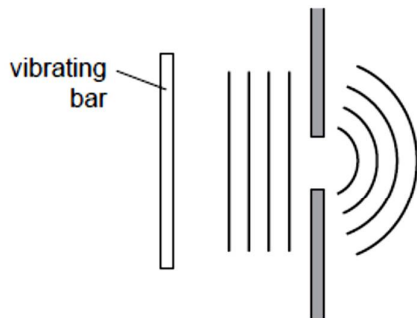


diagram 1

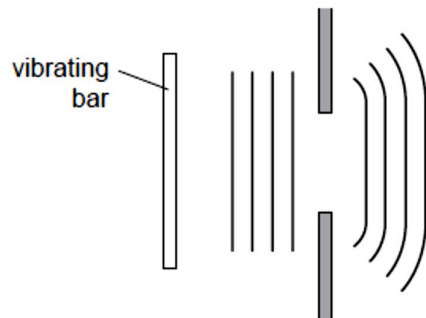


diagram 2

Which action would cause the waves in diagram 1 to produce an emergent pattern closer to that shown in diagram 2?

- A** Increasing the frequency of vibration of the bar.
- B** Increasing the speed of the waves by making the water in the tank deeper.
- C** Reducing the amplitude of vibration of the bar.
- D** Reducing the length of the vibrating bar.

Ans: A

14. An ideal gas in a container of fixed volume 1.0 m^3 has a pressure of $3.0 \times 10^5 \text{ Pa}$ at a